Fenton Nonlinear Wave Suite

Theoretical formulation, solver architecture, equations, and usage

This repository computes steady, two-dimensional, periodic finite-amplitude gravity waves in finite water depth, with optional collinear current. The reference calculation is a stream-function / Fourier collocation method in the Rienecker-Fenton and Fenton numerical-wave tradition: the Fourier basis satisfies the field equation and the bed condition, while Newton iteration solves the nonlinear free-surface boundary conditions and global wave-current constraints.

The central engineering calculation is the nonlinear mapping

$$L = F(H, T, d, U_c),$$

where H is wave height, T is the apparent period in the fixed frame, d is still-water depth, U_c is the imposed collinear current under the selected current convention, and L is the dynamically consistent wavelength. In Fenton's formulation, a steady wave train is fundamentally defined by three length scales, H, d and L. A period can replace wavelength only when the current or wave-speed convention is also specified, because the observed period is Doppler-shifted by current. The program therefore solves wavelength, wave speed, free-surface profile, Fourier coefficients, wave-frame flux, current variables, Bernoulli constant and integral quantities as one coupled nonlinear problem.

Theoretical point	Implementation consequence
The steady-wave model is inviscid, incompressible, irrotational and two-dimensional over a horizontal bed.	The field equation is Laplace's equation; the nonlinear difficulty is concentrated at the free surface.
The Fourier terms are chosen to satisfy Laplace's equation and the bed condition exactly.	The unknowns are solved from free-surface and global constraints, not by fitting a precomputed profile.
If T is specified instead of L, current must also be specified or assumed.	The solver contains explicit Eulerian-current and Stokes/mass-transport current closures.
The method is not a Stokes or cnoidal perturbation expansion.	Increasing N increases numerical resolution rather than changing an analytical order of wave-height expansion.
The wavelength-only and surrogate layers are secondary tools.	They should not be used when velocities, pressures, integral quantities or near-limiting behavior are required.

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1. Repository components

Group	Files	Role in the toolchain
Main GUI solver	`fenton_gui.py`	Python nonlinear solver, GUI, numerical acceptance checks and interval certificates.

Group	Files	Role in the toolchain
Native GUI solver	`fenton_gui.cpp`	Windows C++ GUI with finite-system interval and output-accuracy checks.
Android app	`fenton.apk`	Fenton App for Android.
Wavelength API	`function.py`	Callable wavelength function `L(H,T,d,Uc)`.
Browser solver	`fenton.html`	Self-contained worker-based solver, accuracy checks and TXT/JSON reports.
Standalone precision solver	`fourier.cpp`, `fourier_precision.cpp`	Analytic Jacobian, adaptive continuation, conditioning-based Fourier order and four result/audit files.
Standalone SVD solver	`fourier_carvalho.cpp`	Analytic Jacobian, damped SVD Newton steps, adaptive continuation and three Fenton-style result files.
Tables	`tables.py`	Batch wavelength-table generation.
Pade surrogate	`pade.py`, `pade.bas`	Rational approximation model.
Neural surrogate	`neural.py`, `neural.bas`	Tanh-network approximation model.
Genetic model	`genetic.py`, `genetic.bas`	Explicit symbolic-regression wavelength formula.
Effective-variable model	`effective.bas`	Effective-period/effective-depth wavelength approximation.
Nomograms	`nomogram_plots.py`, `nomogen_plots.py`	Printable design-chart generation.
Nonlinear Excel/VBA solver	`fenton.bas`	`L_fenton` wavelength function and `Fenton_report` numerical verification report, with prescribed Eulerian current.
Spreadsheet deployment	Other `*.bas` modules, `wavelenght.xlsm`	Spreadsheet implementation layer for the respective methods.

The hierarchy is intentional. The stream-function / Fourier solver is the physical reference model. `fenton.bas` solves the coupled nonlinear Fourier system and checks selected integral outputs internally, even when the worksheet requests only wavelength. The separate API, approximation modules and nomograms provide their documented interfaces for wavelength estimation, tabulation or graphical design work.

2. Physical problem and travelling-wave formulation

The physical model is the classical steady progressive wave problem used in Fenton's papers: a homogeneous inviscid fluid, irrotational motion, incompressibility, a flat impermeable bed and a single periodic wave train. The approximation is local in the engineering sense: it is appropriate where the bed can be treated as horizontal over one wavelength and where the wave is non-breaking and changes slowly compared with the periodic motion.

A fixed frame (x, y) is used with x in the propagation direction, $y = 0$ at the still-water level, $y = -d$ at the bed, and $y = \tilde{\eta}(x, t)$ at the free surface. A wave-frame coordinate is introduced by

$$X_{\mathrm{dim}} = x - ct,$$

where c is the wave celerity in the fixed frame. In this frame the surface is steady:

$$\tilde{\eta}(x, t) = \eta(X_{\mathrm{dim}}).$$

The period, wavelength, wave number and celerity are related by

$$c = \frac{L}{T}, \qquad k = \frac{2\pi}{L}, \qquad \omega = \frac{2\pi}{T}.$$

In the program outputs, the reported celerity is the value obtained from the final solved wavelength and the specified period, so the printed geometry remains consistent with this kinematic identity.

Periodicity requires that the surface and the full flow field repeat every wavelength:

$$\eta(X_{\mathrm{dim}} + L) = \eta(X_{\mathrm{dim}}).$$

The specified wave height is the crest-to-trough difference,

$$H = \eta_{\mathrm{crest}} - \eta_{\mathrm{trough}},$$

which becomes, in the nondimensional collocation equations, a difference between the crest and trough ordinates.

Input	Meaning	Typical units
H	Crest-to-trough wave height	m
T	Apparent wave period in the fixed frame	s
d	Still-water depth	m
U_c	Collinear current under the selected convention	m/s
N	Fourier order / number of half-wave intervals	dimensionless

The current U_c is not a harmless correction added after solving a no-current wave. In a steady-wave theory the period observed in the fixed frame is affected by current, and the fixed-frame horizontal velocities require a current convention. The program therefore separates Eulerian current, Stokes/mass-transport current, mean wave-frame velocity and volume flux.

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3. Governing equations before discretization

3.1 Potential-flow equations

The physical velocity components in the fixed frame are obtained from a velocity potential ϕ:

$$u = \frac{\partial \phi}{\partial x}, \qquad v = \frac{\partial \phi}{\partial y}.$$

Incompressibility requires zero divergence,

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = 0,$$

and irrotationality requires zero vorticity,

$$\frac{\partial v}{\partial x} - \frac{\partial u}{\partial y} = 0.$$

Together they imply that the potential satisfies Laplace's equation throughout the unknown water domain:

$$\nabla^2 \phi = \frac{\partial^2 \phi}{\partial x^2} + \frac{\partial^2 \phi}{\partial y^2} = 0,$$

with the vertical range

$$-d \le y \le \eta(X_{\mathrm{dim}}).$$

The bed is impermeable, so the normal velocity there is zero:

$$v(x, -d, t) = 0.$$

3.2 Free-surface kinematic condition

A fluid particle on the free surface remains on the free surface. In the fixed frame this condition is

$$\frac{\partial \eta}{\partial t} + u \frac{\partial \eta}{\partial x} = v \qquad \text{at } y = \eta(x, t).$$

In the wave frame, $U = u - c$ and $V = v$, and the free surface is steady. The same condition becomes

$$U\eta_x = V \qquad \text{on } y = \eta(X_{\text{dim}}),$$

or equivalently

$$(u - c)\eta_x = v \qquad \text{on } y = \eta(X_{\text{dim}}).$$

This is why the stream-function formulation is efficient: in the moving frame the free surface is a streamline, so the kinematic boundary condition can be enforced by making the stream function constant on the unknown surface.

3.3 Free-surface dynamic condition

The pressure is atmospheric and therefore constant on the free surface. Bernoulli's equation in the fixed frame is

$$\frac{1}{2}(u^2 + v^2) + gy + \frac{p}{\rho} + \frac{\partial \phi}{\partial t} = B(t).$$

In the frame moving with the wave, the dynamic condition on the free surface can be written as

$$\frac{1}{2}\left[(u - c)^2 + v^2\right] + g\eta = r,$$

where r uses the still-water-level datum; the corresponding bed-origin constant is $R = r + gd$. This is the main nonlinear boundary condition: velocities are squared and evaluated on the unknown free-surface elevation. The nonlinear solver must therefore find the surface and the velocity field simultaneously.

3.4 Linear dispersion as initialization only

For infinitesimal-amplitude waves without current, linear Airy theory gives

$$\omega^2 = gk \tanh(kd).$$

In wavelength form,

$$\left(\frac{2\pi}{T}\right)^2 = g\frac{2\pi}{L}\tanh\left(\frac{2\pi d}{L}\right).$$

The familiar limiting estimates are

$$L_0 \approx \frac{gT^2}{2\pi}$$

for deep water and

$$c \approx \sqrt{gd}, \qquad L \approx T\sqrt{gd}$$

for shallow water. With a collinear current, the positive intrinsic-frequency branch is

$$\omega - kU_c = \sqrt{gk\tanh(kd)},$$

or equivalently

$$\left(\frac{2\pi}{T} - kU_c\right)^2 = gk\tanh(kd),$$

with branch selection required in adverse-current or near-blocking cases. These formulas are useful for initialization, checking and wavelength-only baselines. They are not the final finite-amplitude theory used by the full solver.

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4. Stream-function formulation

4.1 Definition and boundary interpretation

For two-dimensional incompressible flow in the wave frame, the stream function ψ is defined by

$$U = \frac{\partial\psi}{\partial y}, \qquad V = -\frac{\partial\psi}{\partial x},$$

where $U = u - c$ and $V = v$. Continuity is then satisfied identically. Irrotationality becomes

$$\nabla^2\psi = 0.$$

With the constant in ψ chosen conveniently, the impermeable bed and the moving-frame free surface are streamlines:

$$\psi(x, -d) = 0,$$
$$\psi(x, \eta(x)) = -Q,$$

where Q is the volume flux per unit width in the moving frame. The sign convention follows Fenton's steady-frame formulation: in the wave frame the mean flow is usually in the negative direction under a wave propagating in the positive fixed-frame direction.

4.2 Nondimensional wave-frame coordinates

The numerical formulation uses the wave phase and the mean-level vertical coordinate,

$$X = k(x - ct), \qquad Y = ky, \qquad k = \frac{2\pi}{L}.$$

One wavelength corresponds to

$$X \in [0, 2\pi],$$

and the bed is located at

$$Y = -kd.$$

The nondimensional free surface is written as

$$Y = \zeta(X) = k\eta(X).$$

Define the nondimensional stream function

$$\Psi = \psi\sqrt{\frac{k^3}{g}}.$$

The nondimensional wave-frame velocities are

$$\hat{U} = U\sqrt{\frac{k}{g}} = \frac{\partial\Psi}{\partial Y}, \qquad \hat{V} = V\sqrt{\frac{k}{g}} = -\frac{\partial\Psi}{\partial X}.$$

The Cauchy-Riemann relations between the nondimensional potential and stream function are

$$\Phi_X = \Psi_Y, \qquad \Phi_Y = -\Psi_X.$$

The field equation remains Laplace's equation in nondimensional form:

$$\Psi_{XX} + \Psi_{YY} = 0.$$

The bed and free-surface streamline conditions are

$$\Psi(X, -kd) = 0,$$

and

$$\Psi(X, \zeta(X)) = -Q\sqrt{\frac{k^3}{g}}.$$

The dynamic boundary condition becomes

$$\frac{1}{2}\left(\Psi_X^2 + \Psi_Y^2\right) + \zeta(X) = \frac{rk}{g},$$

or, in residual form with the same mean-level constant r,

$$\frac{1}{2}\left(\Psi_X^2+\Psi_Y^2\right)+\zeta(X)-\frac{rk}{g}=0.$$

The deep-water limiting case is represented by

$$kd\to\infty,$$

with the finite-depth hyperbolic functions replaced by their exponential limits.

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5. Fourier representation and collocation

5.1 Harmonic basis satisfying the field equation

The stream-function / Fourier method is efficient because each vertical basis function is harmonic and satisfies the bottom condition before the nonlinear equations are assembled. A finite-depth nondimensional representation consistent with Fenton's 1988 formulation is

$$\Psi(X,Y)=-\bar{U}^*(Y+kd)+\sum_{j=1}^{N}B_j\frac{\sinh[j(Y+kd)]}{\cosh(jkd)}\cos(jX),$$

where

$$\bar{U}^*=\bar{U}\sqrt{\frac{k}{g}}.$$

At the bed $Y=-kd$, both $Y+kd$ and the hyperbolic sine terms vanish, so

$$\Psi(X,-kd)=0.$$

Differentiating the series gives the nondimensional wave-frame velocity components:

$$\hat{U}(X,Y)=-\bar{U}^*+\sum_{j=1}^{N}jB_j\frac{\cosh[j(Y+kd)]}{\cosh(jkd)}\cos(jX),$$
$$\hat{V}(X,Y)=\sum_{j=1}^{N}jB_j\frac{\sinh[j(Y+kd)]}{\cosh(jkd)}\sin(jX).$$

The equivalent hyperbolic identities are

$$S_j(Y)=\frac{\sinh[j(Y+kd)]}{\cosh(jkd)}=\sinh(jY)+\cosh(jY)\tanh(jkd),$$

$$C_j(Y)=\frac{\cosh[j(Y+kd)]}{\cosh(jkd)}=\cosh(jY)+\sinh(jY)\tanh(jkd).$$

The numerical kernels evaluate these ratios with scaled exponentials. With $a=jY$ and $b=jkd$,

$$S_j(Y) = \frac{e^a - e^{-a-2b}}{1 + e^{-2b}}, \qquad C_j(Y) = \frac{e^a + e^{-a-2b}}{1 + e^{-2b}}.$$

This avoids forming large hyperbolic functions and subtracting nearly equal terms at the bed. In particular, $S_j(-kd) = 0$ and $C_j(-kd) = 1/\cosh(jkd)$. At fixed Y, the deep-water limit is $S_j(Y), C_j(Y) \to e^{jY}$; the finite-depth kernel retains the reflected exponential so that the bed condition remains satisfied.

5.2 Free-surface representation

A symmetric periodic wave can be represented by cosine harmonics. For the surface itself one may write

$$\eta(x) = \sum_{n=0}^{N} a_n \cos(nkx),$$

or in nondimensional phase form,

$$\zeta(X) = \sum_{n=0}^{N} a_n \cos(nX).$$

The implementation follows the Rienecker-Fenton/Fenton numerical approach more directly: it solves for the point ordinates of the free surface at collocation nodes rather than using the surface Fourier coefficients as the primary unknowns. A standard half-wave distribution is

$$x_m = \frac{m\pi}{Nk}, \qquad m = 0, 1, \ldots, N,$$

or in nondimensional form

$$X_m = \frac{m\pi}{N}, \qquad m = 0, 1, \ldots, N.$$

At every node,

$$\zeta_m = \zeta(X_m) = k\eta(X_m).$$

Using the flux convention

$$q^* = \bar{U}^* kd - Q\sqrt{\frac{k^3}{g}},$$

the kinematic free-surface condition can be written as the residual

$$R_m^{(\psi)} = \sum_{j=1}^{N} B_j S_j(\zeta_m) \cos(jX_m) - \bar{U}^* \zeta_m - q^* = 0.$$

The dynamic condition is enforced at the same nodes:

$$R_m^{(B)} = \frac{1}{2}\left[\hat{U}_m^2 + \hat{V}_m^2\right] + \zeta_m - r^* = 0,$$

where

$$r^* = \frac{rk}{g},$$

and

$$\hat{U}_m = -\bar{U}^* + \sum_{j=1}^{N} jB_j C_j(\zeta_m)\cos(jX_m),$$
$$\hat{V}_m = \sum_{j=1}^{N} jB_j S_j(\zeta_m)\sin(jX_m).$$

This separation is important when current is prescribed because the same relative wave form can correspond to different fixed-frame velocities and flux/current conventions.

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6. Dimensionless variables and `solution.res` interpretation

The solver reports quantities using two nondimensional systems: one based on wave number k, and one based on water depth d. The k-based form is natural for Fourier theory, while the d-based form is convenient for engineering interpretation and comparison between depths.

The core nondimensional engineering groups are

$$\hat{H} = \frac{H}{d}, \qquad \hat{L} = \frac{L}{d}, \qquad \hat{c} = \frac{c}{\sqrt{gd}}, \qquad \hat{U}_c = \frac{U_c}{\sqrt{gd}}.$$

Depth and steepness are interpreted using

$$\frac{d}{L}, \qquad \frac{H}{L}, \qquad kd = \frac{2\pi d}{L}.$$

The two common dimensionless periods are

$$\tau_d = T\sqrt{\frac{g}{d}}, \qquad \tau_k = T\sqrt{gk} = \tau_d\sqrt{kd}.$$

The associated depth-based frequency variables are

$$\sigma_d = \frac{2\pi}{\tau_d}, \qquad \alpha_d = \sigma_d^2.$$

The Ursell number, current Froude number and Doppler parameter used for regime interpretation are

$$Ur = \frac{HL^2}{d^3},$$

$$Fr_c = \frac{U_c}{\sqrt{gd}},$$

and

$$D = \frac{U_c T}{L}.$$

The intrinsic wave-current frequency is

$$\omega - kU_c = \omega_{\text{intrinsic}}(k).$$

These nondimensional groups explain why the finite-amplitude wavelength is generally not the linear Airy wavelength:

$$L \neq L_{\text{Linear-Airy}}.$$

6.1 Dimensionless-variable table

Quantity	k-based nondimensional value	d-based nondimensional value
Water depth	kd	$d/d = 1$
Wave length	$k\lambda = 2\pi$	λ/d
Wave height	kH	H/d
Wave period	$T\sqrt{gk}$	$T\sqrt{g/d}$
Wave speed	$c\sqrt{k/g}$	$c/\sqrt{gd}$
Eulerian current	$\bar{u}_1\sqrt{k/g}$	$\bar{u}_1/\sqrt{gd}$
Stokes current / mass-transport current	$\bar{u}_2\sqrt{k/g}$	$\bar{u}_2/\sqrt{gd}$
Mean wave-frame speed	$\bar{U}\sqrt{k/g}$	$\bar{U}/\sqrt{gd}$
Wave volume flux	$q\sqrt{k^3/g}$	$q/\sqrt{gd^3}$
Bernoulli offset	rk/g	r/gd
Volume flux	$Q\sqrt{k^3/g}$	$Q/\sqrt{gd^3}$
Bernoulli constant	Rk/g	R/gd
Momentum flux	$Sk^2/(\rho g)$	$S/(\rho g d^2)$
Impulse	$I\sqrt{k^3}/(\rho\sqrt{g})$	$I/(\rho\sqrt{gd^3})$
Kinetic energy	$T_k k^2/(\rho g)$	$T_k/(\rho g d^2)$
Potential energy	$V_p k^2/(\rho g)$	$V_p/(\rho g d^2)$
Mean square orbital bed velocity	$\langle(u_b - \bar{u}_1)^2\rangle k/g$	$\langle(u_b - \bar{u}_1)^2\rangle/(gd)$
Radiation stress	$S_{xx}k^2/(\rho g)$	$S_{xx}/(\rho g d^2)$
Wave power	$Fk^{5/2}/(\rho g^{3/2})$	$F/(\rho g^{3/2} d^{5/2})$

The Fourier coefficient table is separate: its two columns contain the native stream-function

coefficients B_j and surface coefficients E_j, not two alternative scalings of one quantity. The surface reconstruction is

$$\zeta(X) = \sum_{j=1}^{N-1} E_j \cos(jX) + \frac{E_N}{2}\cos(NX).$$

Thus the terminal cosine amplitude is $E_N/2$. The dimensional surface elevation is $\eta = \zeta/k$; the bed-based ordinate in the standalone surface and flow files is $y/d = 1 + \zeta/(kd)$.

The current definitions used by Fenton can be summarized as

$$\bar{u}_1 = c - \bar{U},$$

and

$$\bar{u}_2 = c - \frac{Q}{d}.$$

The flux variable used in the nonlinear equations is

$$q = \bar{U}d - Q,$$

and the Bernoulli offset is commonly written as

$$r = R - gd$$

when the Bernoulli constant is shifted from a bed-origin formulation to a mean-level formulation.

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7. The `z[i]` state vector and solver description

7.1 Physical meaning of the `z[i]` equations

The `z[i]` equations are not empirical curve-fitting equations. They are the algebraic, nondimensional form of the steady nonlinear free-boundary problem. Solving them means finding a single state vector that simultaneously satisfies:

1. the prescribed wave height and water depth,
2. the specified wavelength or period relation,
3. the selected Eulerian-current or Stokes-current closure,
4. the mean-level convention,
5. the crest-to-trough wave-height condition,
6. the free-surface streamline condition at every collocation node,
7. Bernoulli's equation at every collocation node, and
8. the Fourier representation of a harmonic velocity field.

The unknown vector can be understood as

$$\mathrm{z} = [\text{global scalars} \mid \text{surface ordinates} \mid \text{Fourier coefficients}].$$

The primary physical correspondence in the Fenton-style 1-based state-vector convention is

$$\begin{aligned} z_1 &\leftrightarrow kd, \qquad z_2 \leftrightarrow kH, \qquad z_3 \leftrightarrow T\sqrt{gk}, \\ z_4 &\leftrightarrow c\sqrt{\frac{k}{g}}, \qquad z_5 \leftrightarrow \bar{u}_1\sqrt{\frac{k}{g}}, \qquad z_6 \leftrightarrow \bar{u}_2\sqrt{\frac{k}{g}}, \\ z_7 &\leftrightarrow \bar{U}\sqrt{\frac{k}{g}}, \qquad z_8 \leftrightarrow q\sqrt{\frac{k^3}{g}}, \qquad z_9 \leftrightarrow \frac{rk}{g}. \end{aligned}$$

Block	Indices	Physical/numerical content
Global scalars	`z[1]...z[9]`	kd, kH, period/celerity variables, current variables, flux variable and Bernoulli offset.
Free surface	`z[10]...z[N+10]`	Nodal ordinates $k\eta_m$ from crest to trough.
Fourier modes	`z[N+11]...z[2N+10]`	Harmonic stream-function coefficients B_j.

At the requested order $N = 50$, this indexing corresponds to 110 active 1-based state entries in the Fenton-style indexing convention used by the code. Some ports pack only the independent optimization variables into a shorter zero-based vector; the physical meaning remains the same.

7.2 Current definitions inside the state vector

The program explicitly distinguishes Eulerian current, Stokes/mass-transport current, wave-frame mean velocity and flux. The Eulerian-current variable is connected to celerity and mean wave-frame velocity by

$$\bar{u}_1 = c - \bar{U}.$$

The Stokes-current or mass-transport current is connected to volume flux by

$$\bar{u}_2 = c - \frac{Q}{d}.$$

The wave volume flux convention is

$$q = \bar{U}d - Q.$$

This is not a cosmetic distinction. Prescribing Eulerian current and prescribing Stokes current close different mathematical problems. The residual system therefore selects the active current equation through a current-criterion index.

7.3 Residual equations

The residual vector is the numerical form of the complete nonlinear wave problem. At convergence, every component of

$$\mathrm{F(z)} = 0$$

must vanish. These equations are not optional diagnostics: they are the constraints that define the solved wave. The first eight residuals impose the global wave, current, flux, mean-level and height constraints. The next two blocks are collocation equations imposed at every free-surface node.

For Fourier order N, the half-wave grid contains $N + 1$ collocation nodes,

$$X_m = \frac{m\pi}{N}, \qquad m = 0, 1, \ldots, N.$$

The complete residual system therefore contains

$$8 + (N + 1) + (N + 1) = 2N + 10$$

equations. The requested order $N = 50$ gives 110 equations. If conditioning requires a lower order, the active state and residual system are rebuilt for that order; for example, $N = 32$ gives 74 equations.

Residual group	Number of equations	Unknowns constrained	Physical purpose
Global scalar equations	8	z_1 to z_9 and current selector	Enforce depth, height, period or wavelength, celerity, current, flux, datum and crest-trough height.
Streamline collocation equations	$N + 1$	Surface ordinates and Fourier coefficients	Force every free-surface node to lie on one moving-frame streamline.
Bernoulli collocation equations	$N + 1$	Surface ordinates, velocities and Bernoulli offset	Force constant total head along the unknown free surface.
Complete system	$2N + 10$	Full state vector z	Defines the nonlinear free-boundary wave solution.

The residuals evaluated by the implementation are

$$
\begin{aligned}
r_1 &= z_2 - z_1(H/d), \\
r_2 &= z_2 - H_s z_3^2, \\
r_3 &= z_4 z_3 - 2\pi, \\
r_4 &= z_5 + z_7 - z_4, \\
r_5 &= z_1(z_6 + z_7 - z_4) - z_8, \\
r_6 &= z_{c+4} - \frac{U_c}{\sqrt{gd}}\sqrt{z_1}, \\
r_7 &= z_{10} + z_{N+10} + 2\sum_{i=1}^{N-1} z_{10+i}, \\
r_8 &= z_{10} - z_{N+10} - z_2, \\
r_{9+m} &= \psi_m - z_8 - z_7 z_{10+m}, \\
r_{N+10+m} &= \frac{1}{2}\left[(-z_7 + u_m)^2 + v_m^2\right] + z_{10+m} - z_9,
\end{aligned}
$$

where $m = 0, 1, \ldots, N$ in the two collocation blocks. These expressions describe a finite-depth, period-specified case. With wavelength input, the second residual is instead

$$
r_2 = z_2 - 2\pi H_s, \qquad H_s = H/L
$$

at the current continuation height. The current selector in r_6 is 1 for Eulerian current and 2 for mass-transport current.

7.3.1 Notation used in the residual equations

Symbol	Meaning in the residual system	Practical interpretation
m	Collocation-node index, $m = 0, 1, \ldots, N$	Moves from crest to trough over the symmetric half-wave.
H/d	User-specified relative height	Dimensional input height divided by still-water depth.
H_s	Continuation-step value of the period-form height parameter	For a period-specified problem, it represents the current step of $H/(gT^2)$, so $H_s z_3^2 = kH$ at convergence for that step.
z_{10+m}	Free-surface ordinate at node m	Nondimensional surface elevation $k\eta_m$ solved directly by Newton iteration.
ψ_m	Fourier stream-function contribution evaluated at node m	The sum $\sum B_j S_j(\zeta_m)\cos(jX_m)$, excluding the mean-flow term.
u_m, v_m	Fourier velocity contributions at node m	Used with $-z_7$ to form the total wave-frame velocities in Bernoulli's equation.

Symbol	Meaning in the residual system	Practical interpretation
z_7	Mean wave-frame velocity variable	Appears in both streamline and dynamic boundary conditions.
z_8	Flux-related wave-frame constant	Couples the streamline level to the volume-flux convention.
z_9	Bernoulli offset	The scalar total-head value, after datum shift, that must be the same at every surface node.
z_{c+4}	Current-selected state variable	Selects the active current definition. The selector c is not the wave celerity.

7.3.2 Global residual interpretation The residual equations are listed above in the same order in which they are solved. The table below explains the role of each global residual without changing the formula notation.

Residual	Constraint imposed	What zero residual means
r_1	Height-depth consistency	Since $z_1 = kd$ and $z_2 = kH$, the condition $r_1 = 0$ imposes $kH = kd(H/d)$. The solved nondimensional geometry therefore respects the specified physical H and d.
r_2	Period-form height closure	The current continuation value of $H/(gT^2)$ is consistent with kH and $T\sqrt{gk}$. This is the period-specified counterpart of directly prescribing H/L.
r_3	Period/celerity closure	Since $z_3 = T\sqrt{gk}$ and $z_4 = c\sqrt{k/g}$, $r_3 = 0$ imposes $ckT = 2\pi$, equivalent to $c = L/T$.
r_4	Eulerian-current identity	The state variables satisfy $\bar{u}_1 = c - \bar{U}$.
r_5	Stokes-current and flux identity	The mass-transport current, mean-flow variable and flux-related constant remain mutually consistent. The flux convention cannot drift independently of the velocity field.

Residual	Constraint imposed	What zero residual means
r_6	Prescribed-current equation	The user-specified current is imposed on the selected current variable. The selector c in z_{c+4} chooses the active current criterion and is not the wave celerity.
r_7	Mean-level convention	The discrete free surface has the required zero-mean datum. This removes the otherwise arbitrary vertical translation of the surface.
r_8	Crest-trough wave height	The difference between crest ordinate and trough ordinate equals the solved nondimensional height kH. The solver cannot converge to a nearby wave with the wrong height.

7.3.3 Collocation residual interpretation The two collocation residual families are applied at every node $m = 0, 1, \ldots, N$.

Residual family	Number of equations	Boundary condition represented	What zero residual means
r_{9+m}	$N+1$	Kinematic free-surface condition	The unknown free surface is a streamline in the moving frame. No fluid crosses the solved surface.
r_{N+10+m}	$N+1$	Dynamic free-surface condition	Enforce constant Bernoulli head and atmospheric pressure at the free-surface nodes; between-node accuracy is checked separately.

The collocation residuals are the part of the system that makes the solution fully nonlinear. The surface ordinates are unknown, and the velocities are evaluated on that unknown surface. Newton iteration therefore updates the surface profile, the Fourier coefficients, the celerity/current variables and the Bernoulli offset simultaneously.

Solver consequence	Reason
The wavelength is not obtained from linear dispersion alone.	The period closure is solved together with finite-amplitude free-surface and Bernoulli constraints.
The current affects the nonlinear solution, not only the final reported celerity.	Current variables enter the global residuals and the wave-frame velocity in the boundary conditions.
The Fourier coefficients are not fitted after the surface is known.	They are solved inside the same nonlinear system that determines the surface ordinates.
The mean water level is fixed by the residual system.	The datum equation $r_7 = 0$ removes vertical translation freedom.
The final wave height is enforced directly.	The crest-trough equation $r_8 = 0$ prevents convergence to a nearby wave with the wrong height.

Once $z_1 = kd$ has converged, the wavelength follows directly:

$$kd = \frac{2\pi d}{\lambda}.$$

Thus

$$\frac{\lambda}{d} = \frac{2\pi}{kd}, \qquad L = \lambda, \qquad c = \frac{L}{T}.$$

The residual variable z_4 enforces the same period-celerity closure during Newton iteration; final reports and API diagnostics should still derive the dimensional celerity from the final L/T relation.

7.4 Newton iteration and stabilized linear solve

The residual system is written compactly as

$$\mathrm{F}(\mathrm{z}) = 0.$$

Newton iteration linearizes the residuals about the current iterate:

$$\mathrm{J}(\mathrm{z}^{(m)})\,\Delta\mathrm{z}^{(m)} = -\mathrm{F}(\mathrm{z}^{(m)}),$$
$$\mathrm{z}^{(m+1)} = \mathrm{z}^{(m)} + \Delta\mathrm{z}^{(m)}.$$

The Jacobian entries are

$$\mathrm{J}_{ij} = \frac{\partial F_i}{\partial z_j}.$$

The maintained GUI, browser and standalone solvers assemble analytic derivatives of the global constraints and free-surface equations. The derivatives use the same scaled-exponential basis

as the residual evaluation.

The most sensitive entries involve derivatives of the streamline and Bernoulli residuals with respect to Fourier coefficients, surface ordinates, kd, celerity variables and the Bernoulli offset. Typical derivative families are

$$\frac{\partial R_m^{(\psi)}}{\partial B_n}, \qquad \frac{\partial R_m^{(\psi)}}{\partial \eta_j}, \qquad \frac{\partial R_m^{(\psi)}}{\partial k},$$

and

$$\frac{\partial R_m^{(B)}}{\partial B_n}, \qquad \frac{\partial R_m^{(B)}}{\partial \eta_j}, \qquad \frac{\partial R_m^{(B)}}{\partial r}, \qquad \frac{\partial R_m^{(B)}}{\partial c}.$$

For a dimensional Bernoulli residual,

$$R_m^{(B)} = \frac{1}{2}\left[(u_m - c)^2 + v_m^2\right] + g\eta_m - r,$$

a representative chain-rule derivative is

$$\frac{\partial R_m^{(B)}}{\partial z_j} = (u_m - c)\left(\frac{\partial u_m}{\partial z_j} - \frac{\partial c}{\partial z_j}\right) + v_m \frac{\partial v_m}{\partial z_j} + g\frac{\partial \eta_m}{\partial z_j} - \frac{\partial r}{\partial z_j}.$$

Because near-limiting waves and adverse-current cases can make the Jacobian ill-conditioned, the solvers use stabilized linear algebra. When singular value decomposition is used,

$$\mathrm{J} = \mathrm{U}\Sigma\mathrm{V}^T,$$

and the pseudoinverse Newton correction is

$$\Delta\mathrm{z} = -\mathrm{V}\Sigma^{-1}\mathrm{U}^T\mathrm{F}.$$

The numerical implementations share the governing equations but use different linear and globalization strategies:

Implementation	Linear solve and nonlinear step control
`fourier.cpp` / `fourier_precision.cpp`	Row/column equilibration, LU with iterative refinement, pivoted QR and regularized least-squares fallbacks; trust-region dogleg steps and positive-variable retraction.
`fourier_carvalho.cpp`	Column-equilibrated SVD, relative singular-value cutoff 10^{-15} and residual-decreasing backtracking.
`fenton_gui.py`, `fenton_gui.cpp`, `fenton.html`	Analytic Jacobian, scaled SVD-based correction and safeguarded adaptive continuation.

Implementation	Linear solve and nonlinear step control
`fenton.bas`	Analytic Jacobian, column-equilibrated SVD, backtracking and adaptive height continuation.

Continuation increases wave height from a small-amplitude seed. A failed trial restores the last accepted state and reduces the height increment. Convergence is assessed at the requested final height, rather than at an intermediate wave reached before failure.

7.5 Fourier order and short-period robustness

The requested Fourier order is an upper resolution target. The solvers assess the condition number of the analytic Jacobian after column equilibration:

$$s_j = \max_i |J_{ij}|, \qquad \widetilde{J}_{ij} = \frac{J_{ij}}{s_j}, \qquad \kappa_2(\tilde{\mathrm{J}}) = \frac{\sigma_{\max}}{\sigma_{\min}}.$$

A zero column or singular system is treated as ill-conditioned. If the condition exceeds 10^{11}, the calculation restarts at a lower Fourier order. Starting from 50 modes, the available sequence is

$$50 \to 40 \to 32 \to 25 \to 20 \to 16 \to 12.$$

Each restart solves the original target wave again with the active state-vector size $2N + 10$. It does not reduce the requested physical wave height or relax the final residual tolerance. `fourier_carvalho.cpp` starts at its user-requested order and uses $N_{\text{next}} = \max(12, \lfloor 0.8N \rfloor)$ when the requested order is at least 12; a smaller requested order is not increased.

Short periods such as 3 s and 4 s can make high Fourier orders poorly conditioned for otherwise solvable waves. Adaptive height continuation, scaled basis functions and conditioning-based order selection address this numerical difficulty. A shorter period can also make the specified wave approach or exceed the limiting steady-wave height. Non-convergence is reported if the requested target cannot be verified; an intermediate-height solution is not presented as the requested result.

7.6 Convergence, conditioning and spectral resolution

Residual norms refer to the implemented nondimensional equations, with $M = 2N + 10$:

$$F_{\text{RMS}} = \sqrt{\frac{1}{M}\sum_{i=1}^{M} F_i^2}, \qquad F_{\max} = \max_i |F_i|.$$

Implementation	Intermediate residual limits	Final residual limits
`fourier.cpp` / `fourier_precision.cpp`	$F_{\text{RMS}} \le 10^{-10}$ and $F_{\max} \le 10^{-9}$	$F_{\text{RMS}} \le 10^{-12}$ and $F_{\max} \le 10^{-11}$
`fourier_carvalho.cpp`	$F_{\max} \le 10^{-10}$	$F_{\max} \le 10^{-10}$

Implementation	Intermediate residual limits	Final residual limits
Python/C++ GUI and HTML solvers	$F_{\max} \le 10^{-8}$, together with their step checks	$F_{\max} \le 10^{-10}$, rechecked after post-processing and enclosed independently
`fenton.bas`	$F_{\max} \le 10^{-10}$	$F_{\max} \le 10^{-10}$ after dimensional normalization, plus independent interval checks

The accepted state must also satisfy the finite-value and admissibility checks and the 10^{11} Jacobian-conditioning limit. These are numerical acceptance criteria for a finite Fourier system, not a proof of exact agreement with the continuous free-boundary problem.

The precision driver writes independent diagnostics to `solver.res`:

Diagnostic	Interpretation
Requested and accepted Fourier order	Identifies the resolution actually used.
Fourier order attempts	Lists each attempted order, its acceptance status and condition number.
Adaptive continuation history	Records accepted/rejected height trials for the final order. Earlier order attempts are summarized separately.
Complete state and residual vectors	Provides all $2N + 10$ unknowns and residuals for inspection.
Overall nonlinear convergence	Requires an independently verified final state at the requested height.
Spectral resolution check	Compares the maximum between-node boundary defect with 10^{-9}.

For the precision driver's spectral check, the cosine surface supplies the initial ordinate at each off-grid phase. A safeguarded correction locates the free-surface streamline before the Bernoulli defect is evaluated. The surface output file instead samples the cosine reconstruction directly, so its pressure check need not equal the off-grid diagnostic in `solver.res`.

`fourier_carvalho.cpp` reports the accepted order, maximum residual, Jacobian condition and between-node defect in `solution.res`. Its between-node check evaluates both boundary equations directly on the cosine surface. Consequently its defect is not numerically interchangeable with the precision driver's streamline-corrected diagnostic.

For the standalone drivers described above, a spectral `FAIL` is a separate resolution warning: nonlinear convergence may still be `PASS`. Both statuses must be considered when assessing a result. A failed nonlinear or conditioning check prevents the standalone solvers from writing a new result set; existing files from an earlier run may remain on disk.

7.7 Acceptance of the finite Fourier system and selected outputs

`fenton_gui.py`, `fenton_gui.cpp`, `fenton.html` and `fenton.bas` apply an output-accuracy gate after the nonlinear solve. A result is accepted only when the requested target state satisfies

the conditioning, finite-value, residual, spatial-resolution and output checks. These requirements supplement the nonlinear residual limits in §7.6.

Check	Evaluation	Acceptance requirement
Finite-system residual	Independent interval evaluation of all $2N + 10$ equations at the stored state, reconstructing the dimensionless groups from the physical inputs.	Every residual interval lies inside $[-10^{-10}, 10^{-10}]$.
Interpolated free surface	Both boundary residuals evaluated on the cosine surface at $\max(2048, 32N)$ uniformly spaced phases over one wavelength.	Maximum sampled nondimensional residual no greater than 10^{-6}.
Spectral bed-velocity identities	Compare power and radiation stress with expressions using the independently evaluated absolute mean-square bed velocity.	Each difference fits the output budget below.
Defining integrals	Integrate dimensional velocity and Bernoulli pressure to obtain impulse, kinetic energy, potential energy, radiation stress and power.	Each reported quantity agrees with the finer quadrature within its output budget.
Quadrature refinement	Compare $\max(256, 8N)$ phases and 64 Gauss-Legendre depth points with $\max(512, 16N)$ phases and 96 depth points.	The difference between the two quadratures fits the same output budget.
Selected-output enclosures	Independently enclose nine finite-formula outputs at the stored coefficients.	The worst-case distance from the reported value to its enclosure fits a conservative output budget.

For a dimensional reference quantity q, the comparison budget is

$$\varepsilon(q) = 10^{-7} + 10^{-8}|q|.$$

The absolute term is expressed in the SI units of the quantity being compared. The residual limits are nondimensional. These values are explicit implementation acceptance choices, rather than tolerances prescribed by Fenton for every application.

If the reference is enclosed in $[q_-, q_+]$, the interval check uses

$$e_{\max} = \max(|q_{\text{reported}} - q_-|, |q_{\text{reported}} - q_+|), \qquad e_{\max} \le 10^{-7} + 10^{-8} \min_{q \in [q_-, q_+]} |q|,$$

with outward rounding for the error bound and downward rounding for the budget. The minimum magnitude is zero when the interval contains zero.

The nine enclosed outputs are **wavelength, celerity, impulse, kinetic energy, potential energy, total energy density, radiation stress, wave power and mean-square orbital bed velocity**. These enclosures describe formula evaluation at the stored finite Fourier state. The surface scan and quadrature-refinement comparisons provide additional numerical resolution checks; they are not rigorous bounds over every phase or on quadrature truncation error.

Implementation	Independent enclosure arithmetic
Python	50-digit outward-rounded `Decimal` intervals; the binary64 inputs are interpreted exactly.
C++	Outward-rounded IEEE binary64 intervals using adjacent representable endpoints.
HTML/JavaScript	Outward-rounded IEEE binary64 intervals with explicitly constructed adjacent endpoints.
VBA	Binary64 interval endpoints enlarged by conservative rounding margins, including a margin for subnormal results.

The interval evaluators use bounded trigonometric series independently of the ordinary collocation sine/cosine evaluations. The C++, JavaScript and VBA exponential enclosures use Taylor bounds and range reduction; their square-root endpoint checks validate the initial root estimates.

In the Python, C++ and JavaScript solver objects, successful acceptance sets `converged`, `accuracy_verified` and `finite_system_verified`. The `accuracy_metrics` object records the sampled surface residual and comparison ratios; a ratio at or below one satisfies its budget. `finite_system_certificate` records the residual and output enclosures, budgets and their scope.

`solution_status`	Meaning
`not_solved`	No completed calculation is available.
`verified_numerically`	The nonlinear solution and all implemented acceptance checks passed.
`nonconvergence`	The requested target was not obtained within the solver's admissibility, convergence or conditioning requirements.
`resolution_failure`	The collocation solve completed, but one or more resolution or output-accuracy budgets were exceeded.
`accuracy_failure`	The verification calculation could not be completed, for example because of a numerical exception.

`finite_system_verified` alone does not imply overall acceptance. A finite state can have very small collocation residuals while failing the between-node or integral checks. The GUI and browser display engineering values only for accepted cases, assessing the no-current and imposed-current solutions separately. VBA returns an `Err:` message instead of a wavelength when a check fails; `Fenton_report` exposes the reason and available diagnostics.

7.8 Verification evidence and Lean scope

The Python solver computes accurately for most configurations in the benchmark matrix: 19 of 21 physical cases satisfy all acceptance requirements on both NumPy and Numba. These cases include the default wave, several small period waves, opposing current and both current conventions in the solver core. This is evidence for the tested configurations, not a statistical success rate for arbitrary inputs.

Implementation tested	Benchmark coverage	Result
Python, NumPy and Numba	21 physical cases on each backend; 42 runs.	38 accepted runs covering 19 cases; four accuracy rejections covering two cases.
Native C++ solver core	The same 21 physical cases.	Acceptance decisions and active orders match Python; 21 compared outputs meet the comparison budget for all 19 accepted cases.
JavaScript solver under Node.js	The same 21 physical cases.	Acceptance decisions and active orders match Python; 21 compared outputs meet the comparison budget for all 19 accepted cases.
VBA through a mechanical C++ translation	18 Eulerian-current cases.	Acceptance decisions and active orders match Python; nine compared outputs meet the comparison budget for all 16 accepted cases. Native Excel/VBA execution remains untested.

Independent 80-digit calculations confirm the recorded finite-residual and nine selected-output enclosures for each implementation's tested states. The C++, JavaScript and translated VBA interval arithmetic also pass 911 arithmetic checks. Solver-core verification is distinct from testing every native GUI, spreadsheet host or browser environment.

Two steep-wave configurations are deliberately rejected by the accuracy gate:

H (m)	T (s)	d (m)	Eulerian U_c (m/s)	Active N	Sampled surface residual, approximately
2	3	5	0	32	1.18×10^{-5}
3	4	5	1	40	7.82×10^{-5}

Their collocation residuals are small, but the reconstructed surface and integral comparisons exceed the accuracy budgets. Rejection identifies insufficient numerical accuracy at the available resolution; it does not by itself establish that the physical wave cannot exist.

The companion Python verification package contains five Lean modules:

Module	Formal scope	Theorems
`FentonProduction.lean`	Governing formula, differential and scaling identities.	62
`ProductionAcceptance.lean`	Acceptance rules and contracts for evaluated maxima.	13
`FiniteSystem.lean`	Enclosure arithmetic, pi bounds and finite residual/output-budget implications.	29
`NumericalCertificates.lean`	Exact rational checks on recorded numerical observations.	314
`FiniteCertificates.lean`	Exact integer checks of recorded residual-enclosure and output-budget inequalities.	420
Total	**All modules compiled successfully.**	**838**

The Lean evidence is tied to the source snapshots and recorded states identified by the verification package. Lean checks the mathematical statements, enclosure implications and recorded inequalities. The source-to-model correspondence and runtime interval construction are reviewed and numerically tested; the complete Python, C++, JavaScript and VBA programs are not extracted from Lean or certified as executable programs. The evidence also does not prove existence or uniqueness of an exact nonlinear root, or a continuous water-wave truncation-error bound. This scope is compatible with the demonstrated accuracy of the accepted benchmark calculations.

The verification results in this section apply to the named implementations and their recorded tests. They do not certify the Android binary, separate standalone drivers, wavelength API or surrogate models.

8. Post-processing and engineering outputs

After successful numerical acceptance, the GUI and browser report dimensional engineering quantities. The standalone solvers retain their documented nondimensional output scalings and separate diagnostics. The fundamental dimensional conversions are

$$k = \frac{2\pi}{L}, \qquad c = \frac{L}{T}.$$

This post-processing convention avoids treating wavelength and celerity as independent reported quantities in period-specified runs.

The moving-frame volume flux Q and the wave flux variable q are related by

$$q = \bar{U} d - Q.$$

The mean free-surface level is

$$\bar{\eta} = \frac{1}{L} \int_0^L \eta(x)\, dx.$$

In the usual mean-level convention,

$$\bar{\eta} = 0.$$

The energy calculation separates potential and kinetic components. The total energy is

$$E = E_p + E_k,$$

with potential energy relative to the still-water datum

$$E_p = \frac{\rho g}{L} \int_0^L \frac{\eta^2(x)}{2}\, dx,$$

and kinetic energy

$$E_k = \frac{\rho}{L} \int_0^L \int_{-d}^{\eta(x)} \frac{u^2 + v^2}{2}\, dy\, dx.$$

In the GUI report, the main integral quantities are evaluated with the following dimensional expressions:

$$
\begin{aligned}
I &= \rho(cd - Q), \\
T_k &= \frac{1}{2}(cI - Q\rho\bar{u}_1), \\
V_p &= \frac{1}{2}\rho g\langle\eta^2\rangle, \\
E &= T_k + V_p, \\
F &= c(3T_k - 2V_p) + \frac{1}{2}\langle u_{b,\mathrm{abs}}^2\rangle(I + \rho cd) - 2c\bar{u}_1 I, \\
S_{xx} &= 4T_k - 3V_p + \rho\langle u_{b,\mathrm{abs}}^2\rangle d - 2\bar{u}_1 I, \\
U_{\mathrm{drift}} &= \frac{I}{\rho d}, \\
C_g &= \frac{F}{E}.
\end{aligned}
$$

The integral formulas use the **absolute** bed velocity, including Eulerian current, following Fenton's online *Nonlinear wave theories*, §7. The density factor in kinetic energy is explicit here because I includes density. The invariant and absolute-bed-velocity expressions coincide for an exact steady wave. At a stored finite Fourier state, the invariant formulas supply the main outputs, and the independently evaluated spectral bed-velocity formulas are compared against them within the budgets in §7.7.

$$
\langle u_{b,\mathrm{abs}}^2\rangle = 2r - c(c - 2\bar{u}_1) = \bar{u}_1^2 + \langle(u_b - \bar{u}_1)^2\rangle.
$$

The orbital mean square reported separately by the GUI is not interchangeable with this absolute mean square in F or S_{xx}. When criterion 2 is selected in a supporting solver interface, U_c is the prescribed Stokes current; the integral formulas still use the solved Eulerian current, $\bar{u}_1$.

The field `Cg` is labelled **energy-flux velocity**, F/E. It is an energy-flux-to-energy-density ratio; it is not a general evaluation of the dispersion derivative $d\omega/dk$ for finite-amplitude waves with current.

For the implemented cosine reconstruction, the last mode has amplitude $Y_N/2$, where Y_j denotes the surface coefficient E_j. Consequently the potential energy of that reconstructed surface is

$$
V_p = \frac{\rho g}{k^2}\left(\frac{1}{4}\sum_{j=1}^{N-1} Y_j^2 + \frac{Y_N^2}{16}\right).
$$

Acceptance in the Python/C++ GUI, HTML and VBA solvers requires the residual limits in §7.6 and the resolution and output-accuracy checks in §7.7. A small Newton correction alone is insufficient. The complete target state is checked before accepted engineering outputs are exposed.

The point kernel evaluates the hyperbolic ratios with scaled exponentials to avoid cancellation and overflow at the bed:

$$
S(a,b) = \frac{e^a - e^{-a-2b}}{1 + e^{-2b}}, \qquad C(a,b) = \frac{e^a + e^{-a-2b}}{1 + e^{-2b}}, \qquad a = jY,\ b = jkd.
$$

These quantities should be interpreted with the same current convention used in the solve. The Eulerian-current and Stokes-current closures lead to different fixed-frame flux and drift

interpretations even when the relative wave form is similar.

The orbital mean square bed velocity diagnostic removes the solved Eulerian current:

$$u_{b2} = \left\langle \left(u_{\mathrm{bed}}(t) - \bar{u}_1\right)^2 \right\rangle .$$

The stability diagnostics include the Miche limiting-height estimate,

$$H_{\mathrm{Miche}} = 0.142L \tanh(kd),$$

and the saturation ratio,

$$\text{saturation} = \frac{H}{H_{\mathrm{Miche}}}.$$

The Miche estimate is a practical screening formula rather than the exact limiting wave curve. Near-limiting cases should also be judged by convergence behavior, crest sharpness, Fourier-coefficient decay and the physical plausibility of the branch selected in current.

Output group	Examples	Use
Geometry	L, k, c	Defines the solved wave.
Free surface	η_m, Fourier amplitudes	Crest/trough shape and asymmetry.
Kinematics	u, v, acceleration	Loads, velocity checks and flow fields.
Integrals	I, E, F, S_{xx}	Momentum, energy, power and radiation-stress diagnostics.
Stability	Ur, H_{Miche}, saturation	Practical validity screening.

8.1 Independent pressure and integral checks

Using the mean-level Bernoulli constant r, the dimensional pressure is reconstructed as

$$p = \rho \left[r - gy - \frac{1}{2}\left((u-c)^2 + v^2\right) \right].$$

With $\langle\cdot\rangle$ denoting an average over one wavelength, the independent defining-integral checks include

$$I = \left\langle \int_{-d}^{\eta} \rho u \, dy \right\rangle ,$$

$$S_{xx} = \left\langle \int_{-d}^{\eta} \left(p + \rho u^2\right) \, dy \right\rangle - \frac{1}{2}\rho g d^2,$$

$$F = \left\langle \int_{-d}^{\eta} \left[p + \rho g y + \frac{1}{2}\rho(u^2 + v^2) \right] u \, dy \right\rangle .$$

The kinetic- and potential-energy integrals are defined above. These checks use fixed-frame absolute velocities, including the solved Eulerian current. `MomentumFlux`, a separate moving-frame momentum-flux diagnostic, is not interchangeable with `Sxx` and is not one of the nine interval-enclosed outputs.

8.2 Phase extrema and engineering estimates

The Python/C++ GUI and HTML solvers estimate phase maxima by sampling $\max(257, 8N + 1)$ points on the symmetric half-wave and refining each detected local-maximum bracket with golden-section search. Refinement uses at most 64 iterations or stops when the phase bracket is narrower than 10^{-12} radians. The largest evaluated magnitude is retained. This provides refined numerical estimates, rather than a proof of the global maximum between all sampled points.

Field	Meaning
`u_bed`	Estimated maximum absolute horizontal velocity at the bed, including current.
`u_surf`	Estimated maximum absolute horizontal velocity evaluated on the reconstructed surface.
`w_max`	Estimated maximum absolute vertical velocity evaluated on that surface.
`acc_max`	Estimated maximum absolute material horizontal acceleration evaluated on that surface.
`u_bed_crest`, `u_surf_crest`	Signed horizontal velocities at crest phase; distinct from magnitude maxima.
`extrema_phases`	Phases at which the retained maxima estimates were evaluated.

The material horizontal acceleration used by the GUI and browser is

$$a_x = \frac{\partial u}{\partial t} + u\frac{\partial u}{\partial x} + v\frac{\partial u}{\partial y} = (u - c)\frac{\partial u}{\partial x} + v\frac{\partial u}{\partial y}.$$

This differs from the local acceleration $\partial u/\partial t = -c\,\partial u/\partial x$ written by the standalone flow-field drivers in §11.6.

The engineering bed diagnostics are

$$\tau_{\mathrm{bed,est}} = \frac{1}{2}\rho C_f \, u_{\mathrm{bed}}^2, \qquad C_f = 0.005,$$

$$A_{\mathrm{bed,est}} = \frac{T}{2\pi} U_{b,\mathrm{orb,est}}.$$

Here $U_{b,\mathrm{orb,est}}$ is the estimated phase maximum of $|u_b(X) - \bar{u}_1|$. `ExcursionBed` therefore uses the orbital velocity after removing the Eulerian current. It is an amplitude-based excursion estimate, not direct integration of the nonlinear particle trajectory. Bed shear likewise remains an estimate using a fixed friction coefficient. Miche screening, bed shear and phase extrema are engineering diagnostics outside the nine selected-output interval checks.

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9. Wavelength-only kernels and surrogate formulas

The repository includes approximate layers for speed, spreadsheet deployment and design charts. These formulas are placed here because they are computational derivatives of the same physical mapping $L = F(H, T, d, U_c)$.

9.1 Current-modified linear baseline

For wavelength-only calculations, a common baseline solves

$$(H, T, d, U_c) \mapsto L.$$

With a collinear current, the baseline residual in k is

$$F(k) = \left(\frac{2\pi}{T} - kU_c\right)^2 - gk\tanh(kd).$$

Newton iteration is then

$$k_{n+1} = k_n - \frac{F(k_n)}{F'(k_n)},$$

where

$$F'(k) = -2U_c\left(\frac{2\pi}{T} - kU_c\right) - g\tanh(kd) - gkd\frac{1}{\cosh^2(kd)}.$$

The wavelength is recovered from

$$L = \frac{2\pi}{k}.$$

Without current, the same baseline can be expressed as the fixed-point wavelength equation

$$L_{\mathrm{lin}} = \frac{gT^2}{2\pi}\tanh\left(\frac{2\pi d}{L_{\mathrm{lin}}}\right).$$

9.2 Common surrogate structure

Most reduced-order models correct a physically meaningful baseline:

$$L = L_{\mathrm{base}}C(\mathrm{x}).$$

Typical nondimensional features include

$$\frac{H}{L}, \qquad \frac{d}{L}, \qquad kd, \qquad \frac{U_c}{c}, \qquad \frac{U_c}{\sqrt{gd}}.$$

The implemented engineering feature set includes

$$\begin{aligned}
\text{WaveSteepness} &= \frac{H}{L_{\text{lin}}},\\
\text{RelativeDepth} &= \frac{d}{L_{\text{lin}}},\\
\text{DopplerFactor} &= \frac{U_c T}{L_{\text{lin}}},\\
\text{UrsellNumber} &= \frac{H L_{\text{lin}}^2}{d^3},\\
\text{CurrentFroude} &= \frac{U_c}{\sqrt{gd}}.
\end{aligned}$$

For symbolic-regression and neural workflows, additional logarithmic and current features may be used:

$$x_0 = \ln\!\left(\frac{d}{L}\right), \qquad x_1 = \ln\!\left(\frac{H}{L}\right), \qquad x_2 = \ln\!\left(\frac{H}{d}\right),$$
$$x_3 = \ln(Ur), \qquad Ur = \frac{HL^2}{d^3},$$
$$x_4 = \frac{U_c}{\sqrt{gd}}, \qquad x_5 = \frac{U_c T}{L}, \qquad x_6 = \frac{U_c}{C_0},$$

with the deep-water scale quantities

$$C_0 = \frac{gT}{2\pi}, \qquad L_0 = \frac{gT^2}{2\pi},$$

and the additional features

$$x_7 = \ln\!\left(\frac{H}{L_0}\right), \qquad x_8 = \ln\!\left(T\sqrt{\frac{g}{d}}\right).$$

9.3 Pade/rational surrogate

The Pade surrogate represents a correction with a rational function,

$$P(x) = \frac{a_0 + a_1 x + a_2 x^2 + \cdots + a_m x^m}{1 + b_1 x + b_2 x^2 + \cdots + b_n x^n}.$$

The depth-regime variable is usually

$$\mu = \frac{d}{L}.$$

The regime split used for training and interpretation is

$$\begin{gathered}\mu < 0.05 \quad \text{(shallow)},\\ 0.05 \le \mu < 0.5 \quad \text{(intermediate)},\\ \mu \ge 0.5 \quad \text{(deep)}.\end{gathered}$$

The resulting Pade wavelength expression is

$$L = L_{\text{base}} C_{\text{Pade}}(\mathrm{x}),$$

or explicitly

$$L = L_{\text{base}} \frac{p_0 + p_1 x_1 + p_2 x_2 + \cdots}{1 + q_1 x_1 + q_2 x_2 + \cdots}.$$

9.4 Neural-network surrogate

The neural surrogate is a global tanh-network approximation,

$$\hat{y}(\mathrm{x}) = b_o + \sum_{j=1}^{M} w_j^{(o)} \tanh\left(b_j + \sum_{i=1}^{p} w_{ji}^{(h)} x_i \right).$$

The inputs may be standardized as

$$\tilde{x}_i = \frac{x_i - \mu_i}{\sigma_i},$$

or min-max scaled as

$$\tilde{x}_i = \frac{x_i - x_i^{\min}}{x_i^{\max} - x_i^{\min}}.$$

Training minimizes mean-square loss,

$$\mathcal{L} = \frac{1}{N} \sum_{n=1}^{N} (y_n - \hat{y}_n)^2,$$

and prediction quality is commonly summarized by

$$\text{MAPE} = \frac{100}{N} \sum_{n=1}^{N} \left| \frac{y_n - \hat{y}_n}{y_n} \right|.$$

9.5 Genetic symbolic-regression formula

The genetic symbolic-regression model is an explicit correction over a baseline,

$$L \approx L_{\text{base}} C_{\text{GEP}}(\mathrm{x}),$$

or, more generally,

$$L \approx G(\mathrm{x}).$$

The documented explicit expression is

$$L_{\text{genetic}} = L_{\text{lin}} \left[\exp\left(\frac{345 U_c T}{509 L_{\text{lin}}} \right) + \frac{1052\pi U_c}{1052\pi U_c + 435 g T} - \frac{5 \tanh\left(\frac{U_c T}{L_{\text{lin}}} \ln\left(T\sqrt{\frac{g}{d}} \right) + \frac{2117}{961} \right)}{67 \ln\left(\frac{H}{d} \right)} \right].$$

For readability and implementation checking, it can be decomposed as

$$L_{\text{genetic}} = L_{\text{lin}} M,$$

where

$$\begin{gathered} M = t_1 + t_2 + t_3, \\ t_1 = \exp\left(\frac{345 U_c T}{509 L_{\text{lin}}} \right), \qquad t_2 = \frac{1052\pi U_c}{1052\pi U_c + 435 g T}, \\ t_3 = -\frac{5 \tanh\left(\frac{U_c T}{L_{\text{lin}}} \ln\left(T\sqrt{\frac{g}{d}} \right) + \frac{2117}{961} \right)}{67 \ln\left(\frac{H}{d} \right)}. \end{gathered}$$

9.6 Effective transformed-variable formula

The effective-variable model transforms the input period and depth, then solves a linear dispersion relation with transformed variables. The transformed period and depth are

$$\begin{aligned} T_{\text{eff}} &= T - \frac{93}{943} \left(U_c^2 - \frac{H}{1819 \cdot 708} \right) + \frac{500}{739} U_c, \\ d_{\text{eff}} &= d + \frac{H^{3/4}}{d^{1/4}} + \frac{986}{873} U_c \tanh(H). \end{aligned}$$

Equivalently,

$$T_{\text{eff}} = T + \Delta_T, \qquad d_{\text{eff}} = d + \Delta_d,$$

with

$$\begin{aligned} \Delta_T &= \left(U_c^2 - \frac{H}{1819 \cdot 708} \right) \left(-\frac{93}{943} \right) + \frac{500}{739} U_c, \\ \Delta_d &= \sqrt{\frac{H}{\sqrt{d/H}}} + \frac{986}{873} U_c \tanh(H) = \frac{H^{3/4}}{d^{1/4}} + \frac{986}{873} U_c \tanh(H). \end{aligned}$$

The algebraic identity in the depth transformation is

$$\sqrt{\frac{H}{\sqrt{d/H}}} = \frac{H^{3/4}}{d^{1/4}}.$$

The final effective wavelength is obtained from

$$L_{\text{effective}} = \mathcal{L}_{\text{lin}}(T_{\text{eff}}, d_{\text{eff}}),$$

where

$$\left(\frac{2\pi}{T_{\text{eff}}}\right)^2 = gk\tanh(kd_{\text{eff}}), \qquad L_{\text{effective}} = \frac{2\pi}{k}.$$

The transformed period must remain nonzero:

$$T_{\text{eff}} \neq 0.$$

---

10. Program features

Feature	Description
Fully nonlinear stream-function solve	Solves surface shape, velocity field, wavelength and integral quantities.
Current-aware formulation	Separates Eulerian, Stokes, mean-flow and flux conventions.
Continuation	Adapts height increments and restores the last accepted state after a failed trial.
Fourier order	Requested $N = 50$ in the precision driver; the accepted order is limited by Jacobian conditioning and reported explicitly.
Numerical verification	GUI, HTML and VBA: finite-state interval enclosures, surface-resolution checks and independent output comparisons. Standalone drivers: their documented residual and spectral diagnostics.
Lean evidence	Mathematical identities, finite-system implications and recorded inequalities; scope and benchmark coverage in §7.8.
Short-period calculations	Stable vertical basis and adaptive continuation/order control support tested 3 s and 4 s cases without guaranteeing every height/current combination.
GUI output	Reports accepted geometry, estimated phase extrema, integrals, accuracy diagnostics and breaking-screening quantities.
Wavelength API	Provides callable `L(H,T,d,Uc)` for scripts and tables.
Surrogates	Provides Pade, neural, symbolic-regression and effective-variable approximations.

Feature	Description
Excel/VBA support	`L_fenton` returns an accepted nonlinear wavelength; `Fenton_report` exposes checks and enclosures. Separate modules provide their respective approximations.
Android availability	Fenton App is also available as the installable package `fenton.apk`.
Nomograms	Generates design charts for rapid engineering use.

---

11. Installation and use

11.1 Python GUI

```
python -m venv build_env
build_env\Scripts\activate
python -m pip install --upgrade pip
python -m pip install -U "numpy>=1.20"
python -m pip install -U "numba>=0.57"
python fenton_gui.py
```

Numba is optional. If Numba is unavailable, the solver falls back to the pure-NumPy path. The interval checker uses the standard-library `decimal` module. Tk bindings are required by the GUI; the independent verification package additionally requires its listed audit dependencies.

11.2 Build the C++ GUI executable

```
g++ fenton_gui.cpp -o fenton_gui.exe -O3 -std=c++20 ^
   -fno-fast-math -ffp-contract=off ^
   -march=native -flto=auto -fopenmp -static -static-libgcc ^
   -static-libstdc++ -mwindows -pthread ^
   -lgdi32 -luser32 -lkernel32 -lcomctl32
```

Use a Windows toolchain providing the listed libraries. The interval checker requires IEEE binary64 round-to-nearest arithmetic: do not enable `-ffast-math`, `-Ofast` or floating-point reassociation. The source rejects fast-math builds and checks the arithmetic environment at runtime. Cross-implementation comparisons use explicit tolerances rather than requiring identical last digits.

11.3 Build a Python standalone executable

```
pip install pyinstaller
pyinstaller --onefile --noconsole fenton_gui.py
```

The executable is written to the `dist` directory.

11.4 Wavelength-only use

Use `function.py` when only the wavelength is required by another program or table. The intended callable form is:

```
L(H, T, d, Uc)
```

11.5 Fenton App for Android

The **Fenton App for Android** is distributed as the universal release package `fenton.apk`. It supports Android 6.0 (API 23) and newer and includes ARM64, ARM32, x86 and x86-64 native libraries.

Install `fenton.apk`

1. **Obtain the complete APK.** Download `fenton.apk` directly to the phone, or copy it from a computer into the phone's **Download** folder. Make sure the filename still ends in `.apk`; some messaging and cloud-storage apps rename or incompletely transfer executable files.
2. **Authorize the app that will open the APK.** Android grants local-install permission separately to each source application. Search Android Settings for **Install unknown apps**, **Install unknown applications** or **Installation sources**, select the browser or file manager that will open `fenton.apk`, and enable **Allow from this source**.
3. **Use that same authorized application.** If permission was granted to the file manager, open the phone's file manager, navigate to **Download**, and tap `fenton.apk`. If permission was granted to the browser, open the completed download from that browser. Authorizing a browser does not authorize a file manager, and vice versa.
4. **Approve the Android installer.** Tap **Install** and allow any device security scan to finish. When Android reports that installation is complete, tap **Open** or launch **Fenton Wave Solver** from the app list.
5. **Optionally revoke the source permission.** After installation, **Allow from this source** can be disabled again without affecting the installed application.

Disabling Google Play Protect is normally unnecessary and is not a substitute for the per-source **Allow from this source** permission. Keep Play Protect enabled unless the device administrator or Android itself provides a specific reason to change it.

Troubleshooting installation

Message or symptom	Recommended action
Installation canceled, **Not allowed** or the installer immediately closes	Confirm that **Allow from this source** is enabled for the exact app currently opening the APK. File Manager, Files, Downloads, Chrome and other browsers each have separate permission settings. Return to the APK and try again after enabling it.
The APK is visible but does not open	Move it to the local **Download** folder and open it with the system file manager. Confirm that its name is `fenton.apk`, not `fenton.apk.zip`, and that the transfer completed.

Message or symptom	Recommended action
App not installed while an older Fenton build is present	The installed copy may use another signing certificate or may be newer than the APK. Install a newer signed release, or uninstall the old copy before installing. Uninstalling removes that app's locally stored data.
Invalid package, **Problem parsing the package** or a zero-byte/partial file	Delete the phone copy, download or transfer it again, and compare its size or SHA-256 checksum with the value published for that release.
Installation is blocked on a work, school or supervised phone	A device-administrator policy may prohibit sideloading. Only the administrator can change that policy.
Installation still fails	Check that the device runs Android 6.0 or newer, has adequate free storage, restart it, and retry from the system file manager.

Menu names vary between Android versions and manufacturers. A common path is **Settings > Apps > Special app access > Install unknown apps**, but using Settings search is usually faster.

For advanced diagnosis with Android Platform Tools and USB debugging enabled, run:

```
adb install -r fenton.apk
```

This command prints the precise Android package-manager error when a graphical installer shows only a generic failure message. It is not required for normal installation.

11.6 Standalone precision solver

Compile the `fourier.cpp` source with a C++20 compiler:

```
g++ -std=c++20 -O3 -Wall -Wextra -pedantic fourier.cpp -o fourier
```

Add `-fopenmp` for an OpenMP build where supported. On Windows, use `-o fourier.exe`. The source named `fourier_precision.cpp` uses the same driver and result-file contract; substitute that source filename when compiling it.

The program reads `data.dat` from the working directory or, if absent there, beside the executable. If no file is found, it requests all required case values interactively. Physical inputs have no silent defaults.

`data.dat` contains exactly seven non-empty lines:

Line	Contents
1	Case title
2	Relative wave height $H/d > 0$
3	`Period` or `Wavelength`
4	$T\sqrt{g/d} > 0$ for `Period`, or $L/d > 0$ for `Wavelength`
5	Current criterion: `1` for Eulerian, `2` for mass transport

Line	Contents
6	Signed prescribed current divided by $\sqrt{gd}$
7	`FINISH`

The requested order is 50; the driver selects its initial continuation subdivision internally. Do not add Fourier-order or height-step lines to this file format.

For $H = 3$ m, $T = 9$ s, $d = 5$ m, zero Eulerian current and $g = 9.80665$ m/s^2, use:

```
H=3 m, T=9 s, d=5 m, no current
0.6
Period
12.604274275022739
1
0.0
FINISH
```

Run:

```
./fourier
```

On Windows, run `fourier.exe`. `fourier --help` displays the input contract. When using `data.dat`, results are written beside that file; interactive runs use the working directory.

File	Contents
`solution.res`	Wave geometry, currents, integral quantities and native Fourier coefficients.
`solver.res`	Order selection, conditioning, continuation history, complete state/residual vectors and convergence/resolution checks.
`surface.res`	Surface samples: X_{dim}/d, bed-based elevation y/d, and pressure divided by $\rho g d$.
`flowfield.res`	Bed-to-surface profiles of velocity, local acceleration, velocity gradients and Bernoulli diagnostics.

The `eta/d` label in `surface.res` denotes the **bed-based** ordinate $y/d = 1+\eta/d$: the bed is at 0 and the mean water level at 1. Local accelerations in `flowfield.res` are $\partial u/\partial t$ and $\partial v/\partial t$, not material accelerations including convective terms. The orbital bed-velocity statistic excludes the solved Eulerian current.

Inspect the process exit status and `solver.res` before using results. A nonzero exit status means the requested run did not complete successfully; old `.res` files are not evidence of a successful new calculation.

11.7 Standalone SVD solver

`fourier_carvalho.cpp` retains a separate command-line and file interface:

```
g++ -std=c++20 -O3 -Wall -Wextra -pedantic fourier_carvalho.cpp -o fourier_carvalho
./fourier_carvalho 0.6 Period 12.604274275022739 1 0.0 50 4
```

The arguments are `H_over_d Case Value CurrentCriterion [CurrentValue] N Steps`. `CurrentValue`, when present, follows the current criterion. `N` is the requested Fourier order, from 1 to 512; `Steps` is the initial height subdivision, from 1 to 4096, and continuation subsequently adapts its increments. If `CurrentValue` is omitted, the current is zero.

A discovered `data.dat` takes precedence over positional arguments. This driver reads the title and the same five physical fields as above, followed by **two additional lines**, `N` and `Steps`; a trailing `FINISH` may follow. Use the format for the executable being run: this input is not interchangeable with the precision driver's seven-line file. The SVD driver also retains its separate deep-water branch.

Successful runs write `solution.res`, `surface.res` and `flowfield.res`. Numerical diagnostics are included in `solution.res`; this driver does not produce `solver.res`.

11.8 Browser interface

Open the self-contained `fenton.html` file in a browser and enter H, T, d and Eulerian current U_c in SI units. Calculation runs in a Web Worker. The interface shows accepted engineering values, the active Fourier order, residual bounds and output-accuracy diagnostics. The no-current and imposed-current cases are accepted separately; a rejected column is suppressed without hiding an accepted comparison case.

Use **Full report** for the numerical method, diagnostic ratios and selected-output enclosures. **Download TXT** exports the readable report; **Download JSON** exports the input, solver statuses, Fourier state and verification data. JSON may contain diagnostic states for rejected cases; use the acceptance flags before treating any values as engineering results. **Clear output** cancels a pending calculation and clears the displayed results.

JavaScript uses binary64 arithmetic, so final machine-level digits can vary between engines. The recorded numerical comparison in §7.8 executes the solver under Node.js; it does not assert identical execution in every browser.

11.9 Excel/VBA nonlinear wavelength and verification report

Import `fenton.bas` into a **standard VBA module** through the Visual Basic Editor's **File > Import File** command, and save the workbook as `.xlsm`. The module uses SI inputs, $g = 9.80665$ m/s^2 and, for the internal integral checks, $\rho = 1025$ kg/m^3. Its prescribed current is Eulerian.

The wavelength function retains this interface:

```
L_fenton(H, T, d, [U], [UseHomotopy])
```

`U` defaults to zero. `UseHomotopy` is retained for worksheet compatibility; adaptive height continuation is always applied. A successful call returns wavelength in metres. A failed nonlinear solve, inadmissible state or failed accuracy check returns an `Err:` text message, rather than a partial or unverified wavelength.

```
=L_fenton(3,9,5)
=L_fenton(3,9,5,1)
=Fenton_report(3,9,5,1)
```

Use semicolons instead of commas if required by Excel's regional settings. `Fenton_report(H,T,d,[U])` returns a multiline text report; enable **Wrap Text** in its cell.

It includes the acceptance outcome, active order, conditioning, residual and comparison diagnostics. For accepted cases it also lists the nine checked outputs and their enclosures. Each call performs its own solve and verification, so extensive worksheet tables can take appreciable time to recalculate.

For the default wave $H = 3$ m, $T = 9$ s and $d = 5$ m, representative accepted wavelengths are:

Eulerian current (m/s)	Wavelength (m), approximately
0	68.706761529
1	78.827222116

The VBA numerical comparison uses a mechanical C++ translation of the module, with independent high-precision enclosure checks. Native Excel compilation and execution have not been run in that verification environment.

11.10 Python and Lean verification

Run the Python solver's built-in acceptance/rejection checks with:

```
python fenton_gui.py --self-test
```

The companion verification package contains its own source snapshots, dependency requirements, manifests, numerical audit and Lean build configuration. From that package directory, its reproduction sequence is:

```
python -m pip install -r requirements.txt
python audit.py
python check_intervals.py
python check_report.py
python inputs/fenton_gui.py --self-test
python build_certificates.py
python build_finite_certificates.py
lake update
lake exe cache get
lake build
```

The package pins Lean 4.34.0 and Mathlib commit `5ed2965256430c3649e86755f9576b54eca72435`. The audit checks the source identities before generating recorded certificates. Successful reproduction requires both the numerical tests and all five Lean modules to pass. Deliberate rejection of an under-resolved case is a passing acceptance test when that rejection is the expected outcome. Read §7.8 for the exact scope of the resulting evidence.

---

12. Validity and numerical cautions

Condition	Practical implication
Non-breaking steady waves	The formulation is for a periodic wave of permanent form; breaking, strong turbulence and air entrainment are outside the model.

Condition	Practical implication
Irrotational flow	Strong shear, vorticity, boundary-layer dominance or rapidly varying currents require a different model.
Horizontal-bed assumption	The steady solver is a local horizontal-bed theory; shoaling over variable bathymetry requires an additional propagation/shoaling model or a local approximation.
Very long waves	Fourier coefficients become broad-banded near the solitary-wave limit. Inspect coefficient decay and between-node defects; a larger requested N is useful only while the system remains well-conditioned.
Near highest waves	Sharp crests degrade convergence and conditioning; inspect the achieved height, active Fourier order, residuals and free-surface shape.
Short periods	For fixed height and depth, a shorter period can produce a much steeper wave. Automatic order reduction addresses conditioning, but cannot establish the existence of a steady solution.
Resolution and acceptance	Small collocation residuals alone do not establish between-node accuracy. GUI/HTML/VBA results are rejected when their accuracy budgets fail. Standalone spectral `FAIL` and nonlinear `PASS` remain separate diagnostics under their documented contracts.
Finite-system verification	Enclosures concern equation and output evaluation at the stored coefficients; they do not include physical-input uncertainty, an exact-root error bound or continuous Fourier-truncation error.
Estimated extrema	Refined phase searches estimate maxima. Bed shear and excursion are engineering estimates with the definitions in §8.2.
Comparing implementations	Match height, depth, period or wavelength, current definition, current value, gravity, active order and tolerances. Differences in quadrature and floating-point linear algebra can prevent exact numerical equality.
Opposing current	Wave blocking, multiple branches or no physical branch may occur; the chosen root must be checked.
Current criterion	Do not mix Eulerian-current and Stokes/mass-transport current definitions.

Condition	Practical implication
Surrogate use	Do not extrapolate outside the training/calibration range. Surrogates are wavelength-estimation tools, not replacements for the nonlinear kinematic solution.
Units	GUI inputs use coherent SI units; standalone `data.dat` inputs are the dimensionless groups specified in §11.6.

Fenton's theory comparisons give the following practical hierarchy. Fifth-order Stokes theory is useful for shorter waves, cnoidal theory for sufficiently long waves, with the Ursell number often used as a regime guide. The Fourier approximation method is the preferred reference method when high accuracy is required over a broad range of depths and heights, especially for velocities, pressures, integral quantities or waves close to the boundary of the perturbation theories.

The nonlinear solver is therefore the preferred method when free-surface shape, velocities, accelerations, energy, radiation stress or near-breaking behavior matter. The wavelength-only kernels, symbolic formulas, Pade model, neural model and effective-variable model are appropriate for fast wavelength estimates only when their calibration range is respected.

---

13. Dedication

This work is dedicated to **Dr John D. Fenton** (1945-), Australian civil engineer, hydraulic engineer, and applied mathematician, in recognition of his distinguished and continuing scientific contributions to mathematical hydraulics, computational fluid mechanics, and nonlinear water-wave theory. Further information, publications, technical resources, and software by Dr Fenton are available on his official website: www.johndfenton.com.

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14. References

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